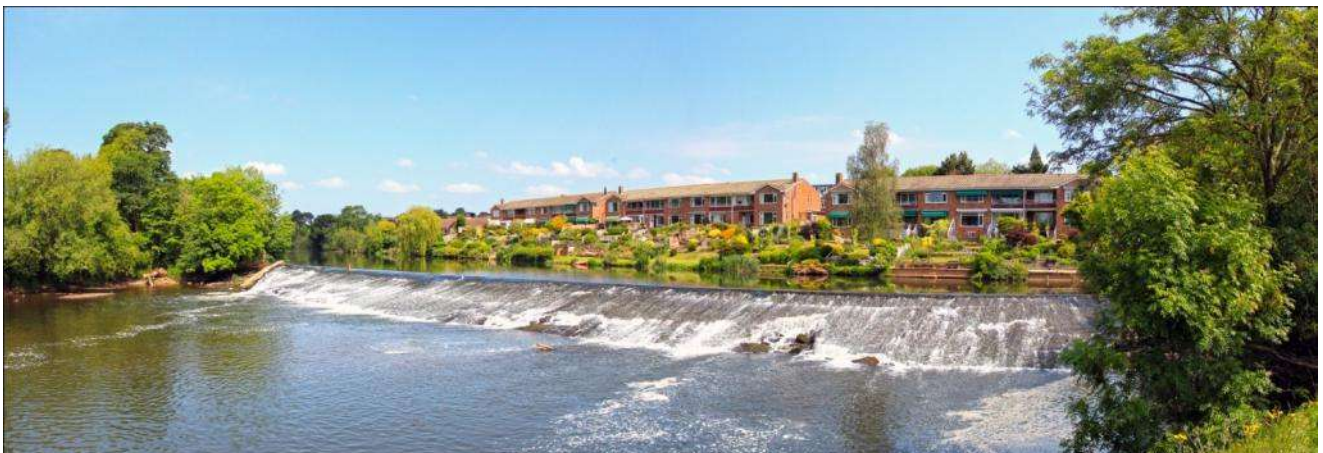

St James' Weir Assessment Report

Document: R10 Version: 2

South West Fish Passes

Environment Agency

28 May 2013





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28 May 2013

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Document history

St James' Weir Assessment Report

South West Fish Passes

Environment Agency

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Appendix

Appendix A **Cost Estimates**

- A.1 Construction Cost Estimate for Do Minimum Option
- A.2 Construction Cost Estimate for Comprehensive Repair Option

1 Introduction

St James' Weir lies on the River Exe in Exeter at Grid Reference SX 9303 9094. The weir, which is also known as Salmon Pool Weir, was constructed in the 14th century by the Courtenays to raise the river level for the St James' Mill Leat that drove their mills further downstream in Countess Wear¹. The weir is an important feature within Exeter and has significant archaeological, recreational and amenity value.

The weir forms the tidal limit of the River Exe and is the first barrier that migratory fish encounter on their way to spawning grounds upstream on the River Exe and its tributaries. There is an existing baulk fish pass on the right side of the weir which was refurbished by Environment Agency Field Services approximately 10 years ago, but is not considered effective for fish passage.

At present no one claims ownership of the weir, however it is understood that the River Exe Foundation has submitted an affidavit to take ownership of the weir between the right bank and existing baulk fish pass.

In recent times the weir has reportedly started to disintegrate prompting fears that it may collapse. As the weir serves no flood defence purpose², the Environment Agency has no formal remit to implement repairs.

The Environment Agency South West Region is delivering a programme of improvements to fish passage to meet its obligations under the Water Environment (Water Framework Directive) (England and Wales) Regulations 2003 which aims to improve the ecological status of our rivers. One of the sites targeted for improvement is St James' Weir, where the Environment Agency is considering constructing a Larinier fish pass and eel pass which have been designed by Halcrow Group Ltd (Halcrow) for the Westcountry Rivers Trust. If constructed the pass would allow improved fish passage past the weir for a wide range of fish species.

The construction of the new fish pass would be a significant financial investment (construction cost estimated to be around £500k). A collapse of a significant part of the weir would make the fish pass redundant and if this was to occur in the short term, would make investment in the new fish pass an ineffective use of public funds. The aim of this study is to:

- Carry out a condition assessment of the weir;
- Estimate if/when a significant breach of the weir can be expected;
- Identify and cost options for repairing the weir; and
- Consider the consequences of the weir collapsing and/or being removed.

Left and right bank referenced below refer to the true left and right when looking downstream.

¹ Exeter Memories website (<http://www.exetermemories.co.uk>)

² St James' Weir – Appraisal of Damage (Halcrow Group Ltd, 2012)

2 Weir Condition Assessment

2.1 Weir Geometry

St James' Weir is located on a sharp right hand bend in the River Exe. The weir is overlooked by Old Abbey Court Road residential properties located on the left bank upstream of the weir. The entrance to St James' Mill Leat is located on the left bank immediately upstream of the weir with flow into the leat being regulated by a fixed concrete control structure. A footbridge connects Salmonpool Lane with Duck's Marsh playing fields on the left bank. The right bank is a mix of open recreational land and deciduous woodland. It is understood the right bank is owned by the Environment Agency. See Figure 2-1 below.

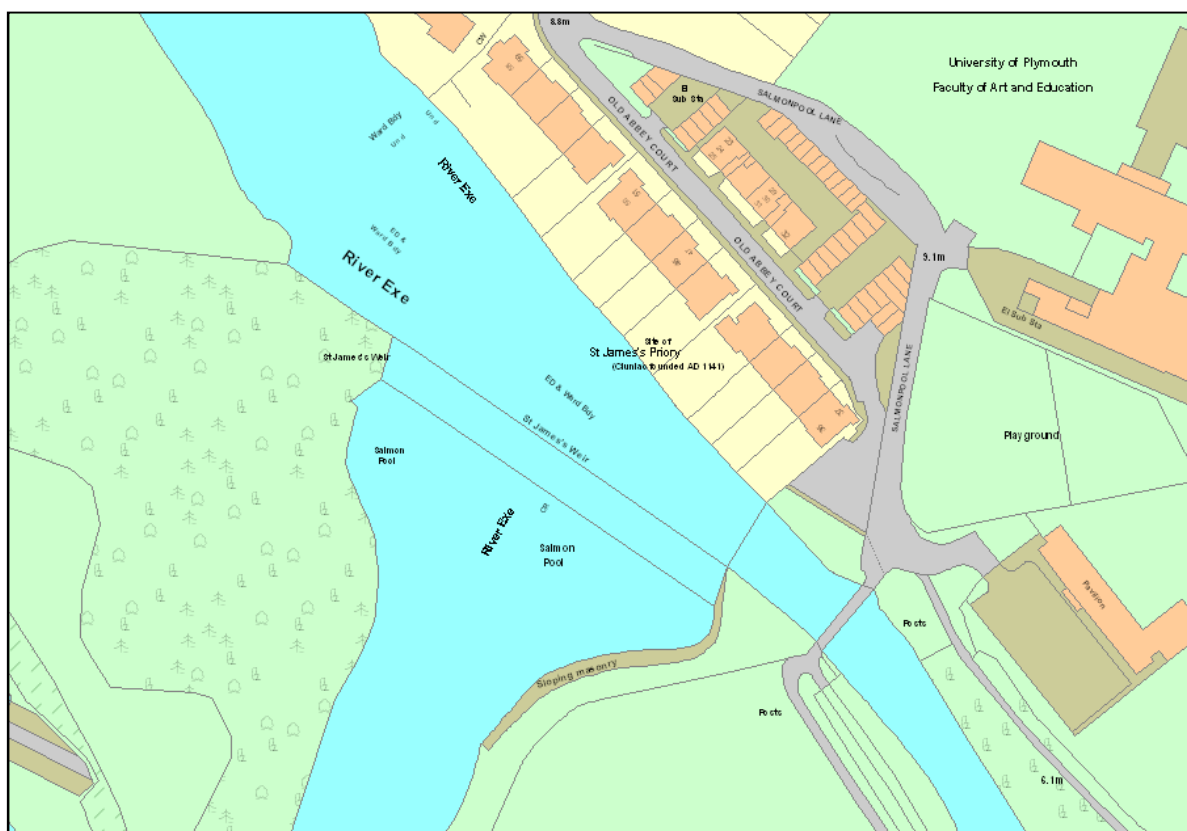


Figure 2-1: St James' Weir

The weir is 97m long, 10-12m wide and the weir crest level varies from 3.32mAOD to 3.48mAOD. The weir has a sloping glacis, the base of which has a level of 0.79mAOD to 1.29mAOD. The river bed level immediately downstream of the weir has a level of -0.24mAOD to 0.66mAOD. A large deposit of silt has built up behind the weir (on the upstream side) and the bed level here is at most 0.8m below the weir crest level.

Sections through the weir are shown below in Figure 2-2 and Figure 2-3.

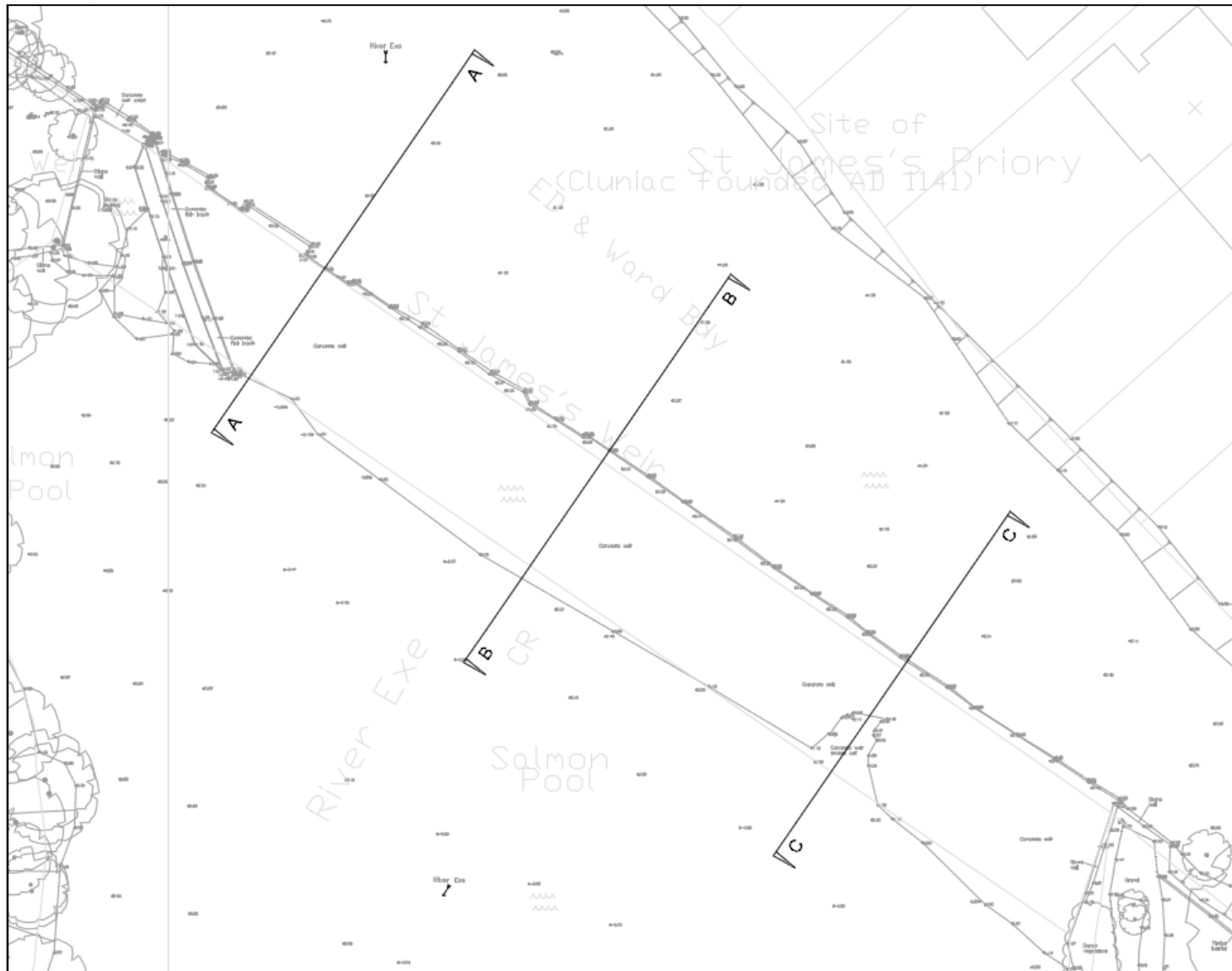


Figure 2-2: Cross Section Locations

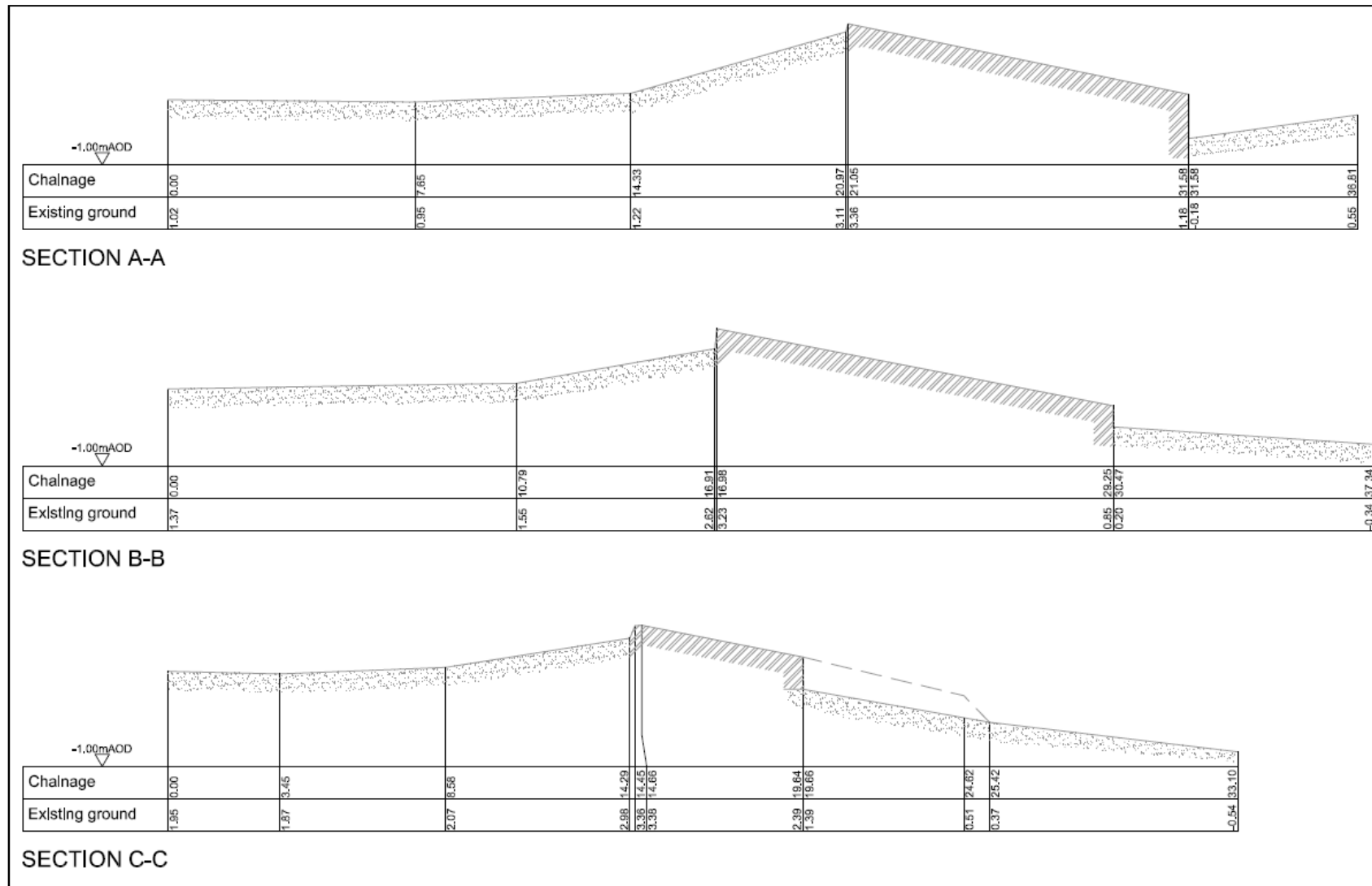


Figure 2-3: St James' Weir Cross Sections

In 1996 a ground investigation was carried out at the weir comprising two boreholes in the right bank, a dynamic probe through the weir surface adjacent to the baulk pass and two dynamic probes in the river bed downstream of the weir.

Ground conditions in the river channel are typically gravel alluvium overlying Breccia bedrock. Bedrock level in the river channel is located at -0.54mAOD to +0.21mAOD. It is expected that the weir is founded on bedrock.

2.2 Weir Construction

A common form of construction of weirs of this age is oak stakes placed six feet apart which were tied with elm branches (1" diameter tied in bundles). Stone was laid in layers between the elm bundles which are designed to swell up when wet³.

A dynamic probe taken through the weir surface adjacent to the baulk pass suggests the stone and timber weir surface is 0.6m thick and is underlain with alluvial material and founded on bedrock.

The weir glacis, crest and upstream face of the weir have been covered with a 100mm to 300mm thick layer of concrete at some point in the past.

Timber stakes and rectangular facing stones were observed during the weir inspection. The weir stones are typically rectangular blocks with plan dimensions ranging from 300mm x 150mm to 300mm x 500mm. The stone is reportedly (Ian Cook – River Exe Foundation) locally sourced Heavitree stone. See Figure 2-4 and Figure 2-5 below.

³ St James' Weir Fish Pass Appraisal Report (Halcrow Group Ltd, 1997)

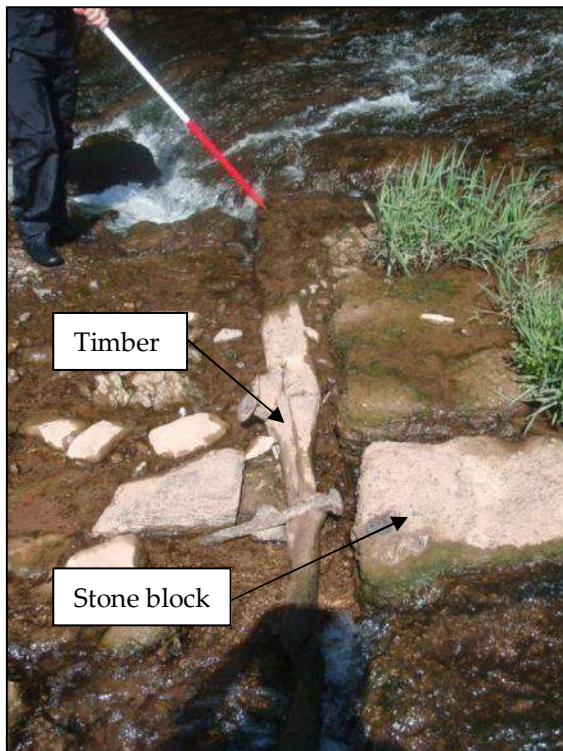


Figure 2-4: Original weir construction (right side of weir)

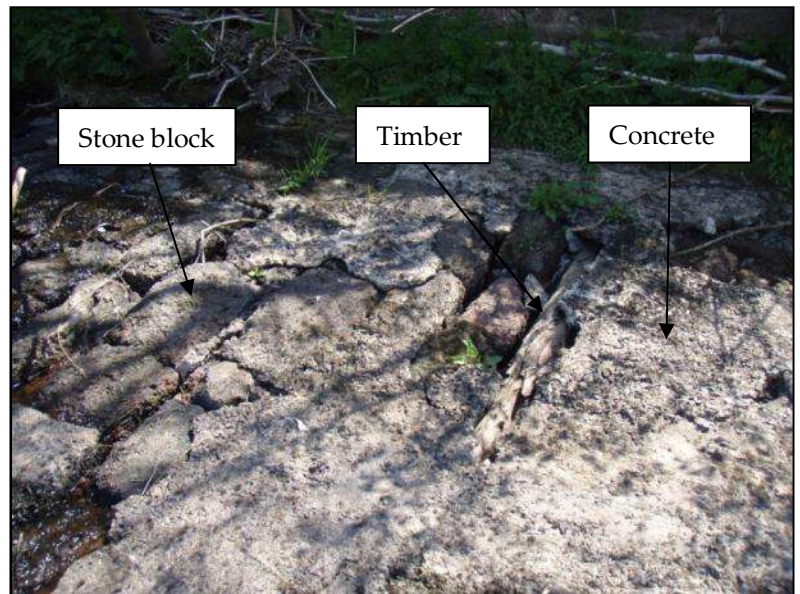


Figure 2-5: Original weir construction (left side of weir)

2.3 Weir Survey

In May 2013 Halcrow undertook an inspection of the weir to assess its condition and record signs of deterioration. An experienced four man team undertook the inspection using a safe system of work. River flow and tide level were low on the day of the inspection allowing surveyors to traverse the upstream and downstream sides of the weir.

The aim of the inspection was to identify and record defects in the weir which could affect its structural integrity. The most common problems affecting masonry weirs are scour at the toe, damage to the weir surface, mortar loss and leakage.

Areas where significant defects were identified are shown in Figure 2-6 and Figure 2-7 below and a detailed description of each defect is provided in section 2.4 below.

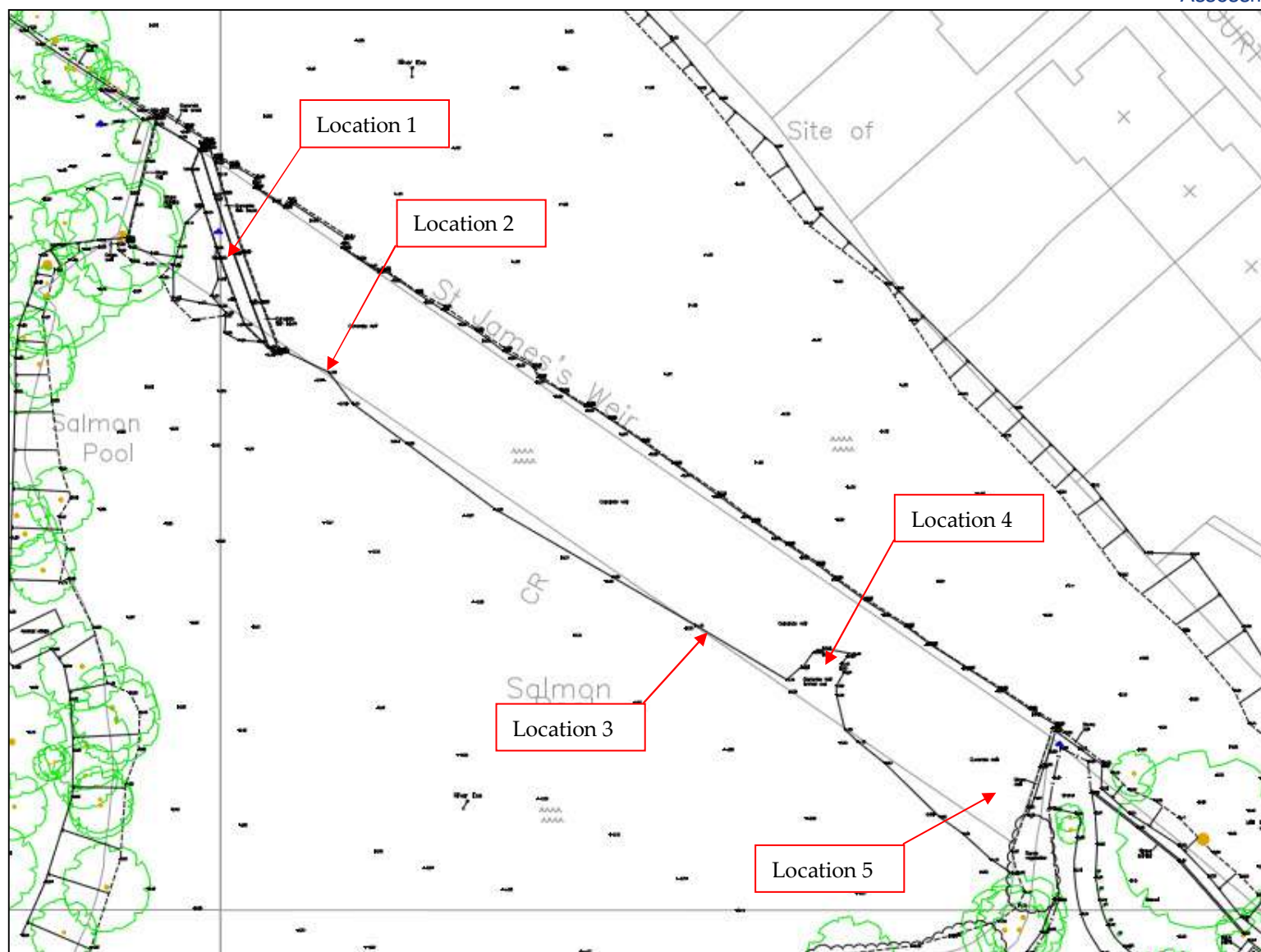


Figure 2-6: Defect location plan



Figure 2-7: Defect locations

2.4 Weir Condition Assessment

In the following weir condition assessment, timescales are referenced in the following way:

Short term: 0 to 10 years

Medium term: 10 – 20 years

Long term: > 20 years

By comparing photos of the weir taken in 2013 with historical photos from 2003 and 1996 it is possible to assess the rate at which different parts of the weir are deteriorating.

2.4.1 Location 1

Location 1 is the area around the existing baulk fish pass. Significant leakage through the weir surface and under the baulk pass structure was observed. Sections of the concrete apron on the downstream side of the baulk appear to have broken away. The original weir stone is exposed and any mortar that was laid between the stones has been eroded. See Figure 2-8 and Figure 2-9 below.

The concrete baulk has been laid onto the weir surface without any kind of key into the main body of the weir which has allowed a flow path to form underneath.

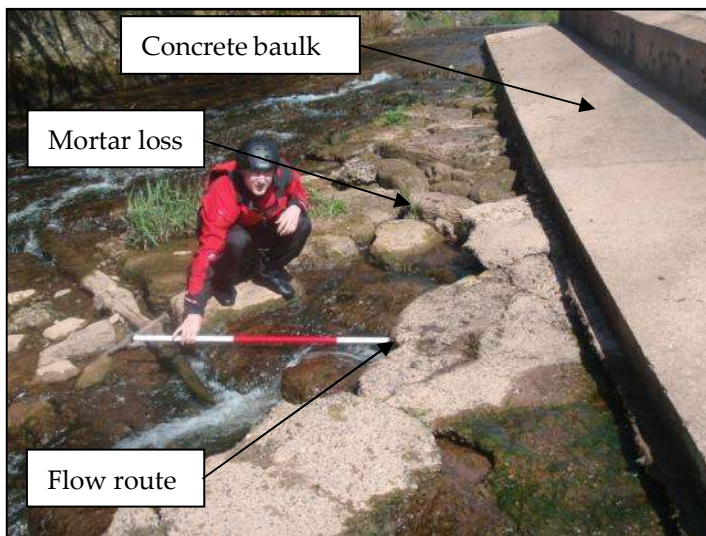


Figure 2-8: Original weir construction (right side of weir)

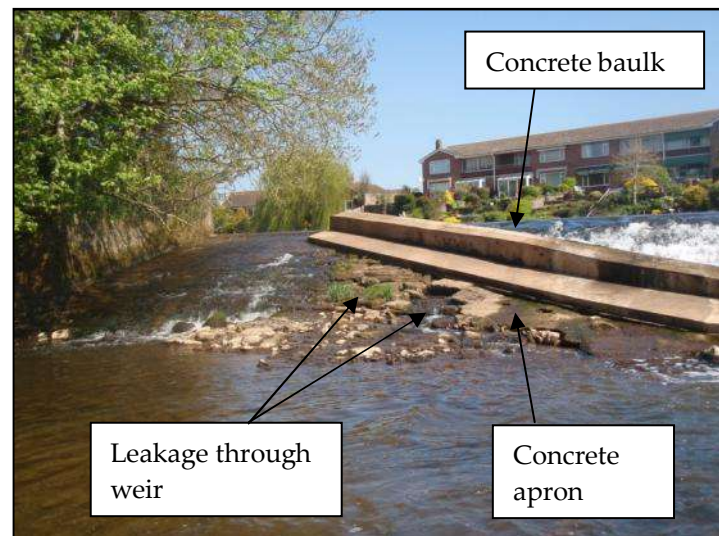


Figure 2-9: Original weir construction (left side of weir)

The end of the baulk pass has also been scoured and is undercut by approximately 1m. The baulk structure is formed from reinforced concrete (Tony Seaward, Environment Agency Field Services, pers comm) and has the structural capacity to cantilever out over the river bed even with further undermining. It is expected that the baulk will remain intact in the short and medium term but further undercutting and loss of weir toe at the baulk entrance is a threat to the integrity and efficacy of the pass in the long term.

Whilst leakage and mortar loss in the weir at this location is considered to be a defect, a photo from 1997 (see Figure 2-10 below) taken before the baulk improvement works were carried out, shows the weir to be in similar condition at that time. Therefore, the weir in this location does not appear to be deteriorating at a noticeable rate and does not in itself present a threat to the integrity of the weir itself in the short or medium term.



Figure 2-10: Location 1 in 1996

2.4.2 Location 2

Location 2 is the section of weir toe to the left of the baulk. An approximately 8m wide x 1m wide section of the weir toe has broken away in this location, see Figure 2-11 and Figure 2-12 below. The most likely cause of this failure is progressive scour under the weir toe caused by turbulence in the weir pool. The broken section of weir is a mix of stone and concrete. It remains in place and has probably provided some protection to further scour and erosion.

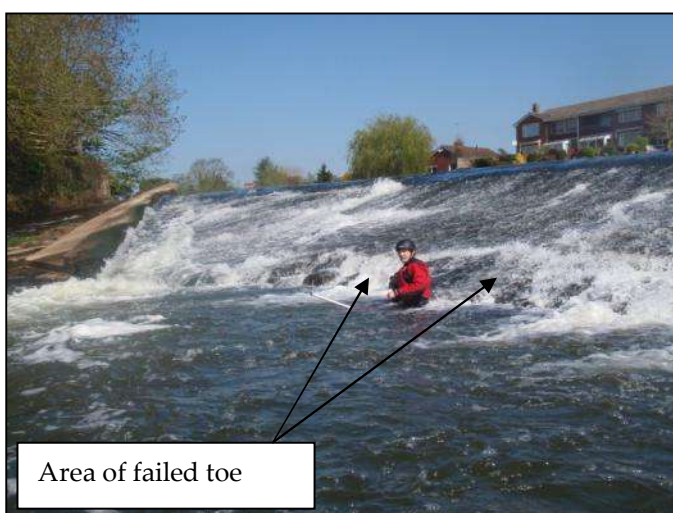


Figure 2-11: Original weir construction (right side of weir)

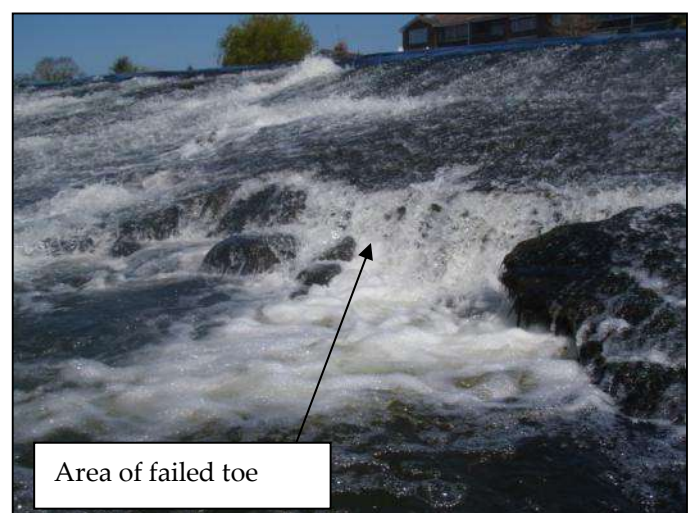


Figure 2-12: Original weir construction (left side of weir)

Historical photos show this failure originally occurred at some time between 1996 and 2003, see Figure 2-13 and Figure 2-14 below. The failure at this location

does not appear to have significantly worsened since 2003 and is therefore not considered to be a threat to the integrity of the weir in the short or medium term.



Figure 2-13: Location 2 in 2003 (Mott MacDonald)

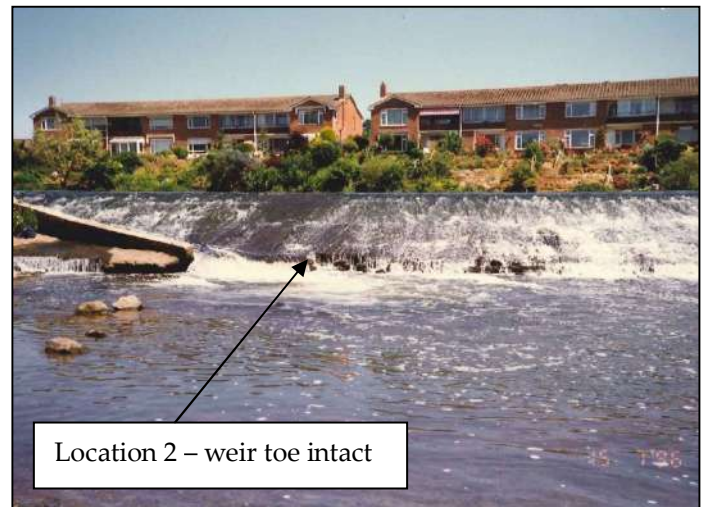


Figure 2-14: Location 2 in 1996

2.4.3 Location 3

Location 3 is section of weir toe approximately two thirds of the way across the weir from the right bank. A section of the weir toe approximately 12m long x 3m wide has collapsed in this location leaving a vertical cut in the weir glacis and a mix of stone and concrete rubble below, see Figure 2-15 and Figure 2-16 below. The remaining vertical face is undercut by up to 0.5m. The collapse was most likely due to undercutting from downstream caused by turbulence in the weir pool.

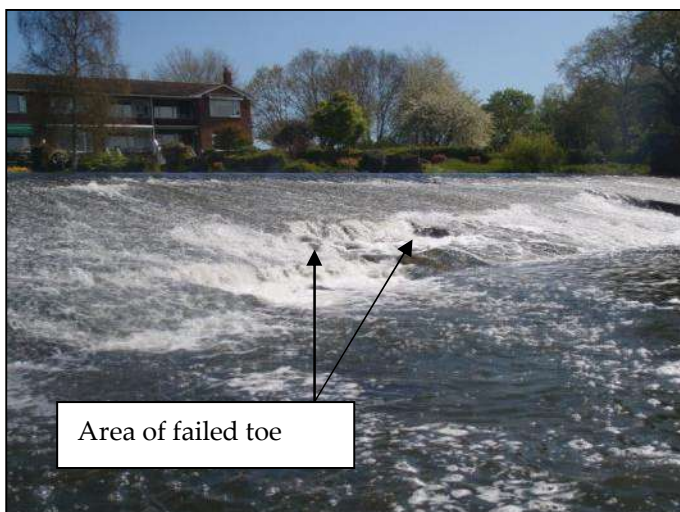


Figure 2-15: Location 3

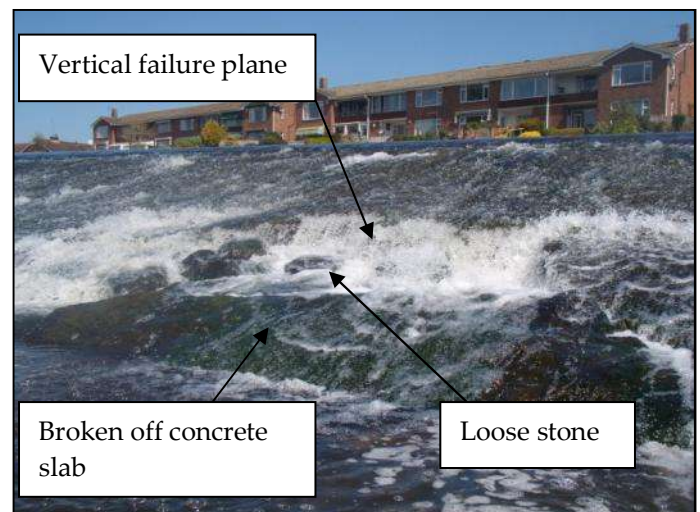


Figure 2-16: Location 3

As at location 2, the collapsed section is formed from a mix of concrete and stone, and this has probably helped to prevent the hole rapidly increasing in size.

It is expected that any further scour under the vertical face of the hole will lead to a progressive increase in the size of the hole, which will migrate towards the weir crest over time.

From historical photos, see Figure 2-17 and Figure 2-18 below, damage in this area was not apparent in 1996 but initial failure was evident at this location in 2003.



Figure 2-17: Location 3 in 2003 (Mott MacDonald)



Figure 2-18: Location 3 in 1996

A comparison of the 2003 photo and 2013 photo, see Figure 2-19 and Figure 2-20 below, shows that the damage in this area has not significantly worsened in the last 10 years.



Figure 2-19: Location 3 in 2003 (Mott MacDonald)



Figure 2-20: Location 3 in 2013

It is considered that the damage in this area is not a threat to the integrity of the weir in the short term, however further scour and progressive failure is expected in the medium term which could ultimately result in a breach in the weir.

2.4.4 Location 4

Location 4 is the most significant area of damage of the weir. The damage comprises a large horseshoe shaped hole in the weir glacis. The edge of the whole is vertical and around 1.0m high. The vertical face is undercut by approximately 0.5m leaving a 4m width of weir between the back of the hole and the weir crest. In the base of the hole is a mixture of stone and concrete rubble. The hole extends approximately halfway through the weir structure, see Figure 2-21 and Figure 2-22 below. It would appear that the failure was initiated by progressive undermining of the toe from downstream due to turbulence in the weir pool. This led to the subsequent formation of a void under the glacis, leading to collapse of the glacis.

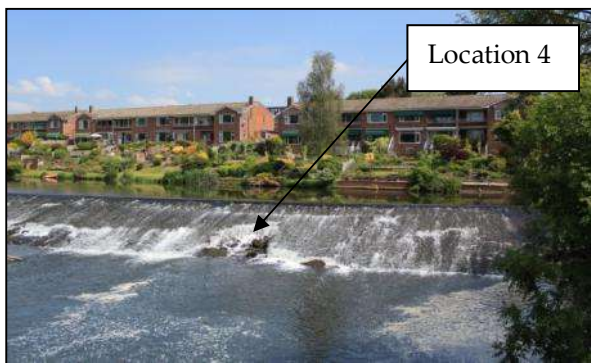


Figure 2-21: Location 4



Figure 2-22: Location 4

Historical photos, shown in Figure 2-23 and Figure 2-24 below, show the first signs of damage in this location in 1996. By 2003 the damage does not appear to have significantly worsened. However by 2013 the hole has grown significantly, extending through half of the weir.



Figure 2-23: Location 4 in 1996 (Halcrow Group Ltd)



Figure 2-24: Location 4 in 2003 (Mott MacDonald)

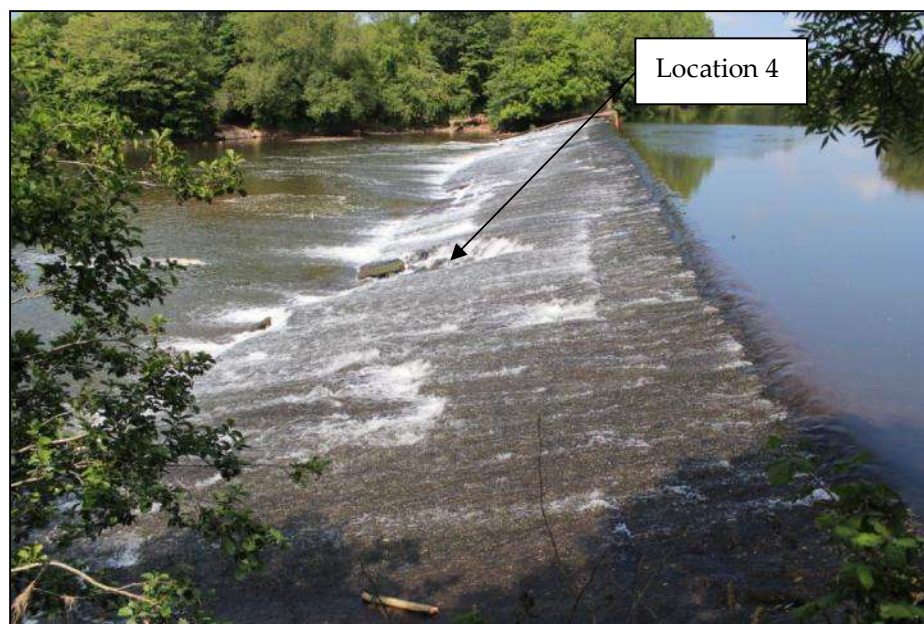


Figure 2-25: Location 4 in 2013



Figure 2-26: Location 4 in 2013

The rate at which the hole continues to grow will depend on the number of flood events occurring in forthcoming years as most damage is likely to be caused during high flows. If the hole continues to grow at the same rate as it has over the last 10 years, then a full breach of weir can be expected by 2023.

Once the main weir crest is breached and the alluvial material in the core of the weir exposed, it is expected that the breach would rapidly grow in size with a cycle of scour of the core material followed by collapse of the weir surface progressively widening the breach.

Once a significant breach in the weir has formed, water levels upstream of the weir would drop causing the St James' Mill Leat to run dry. If a Larinier fish pass has been constructed, this would become redundant.

2.4.5 Location 5

Location 5 is the section of the lower half of the weir adjacent to the left bank. Significant leakage through the weir surface was observed. Sections of the concrete have broken away and stone blocks underneath are missing. In places, the original weir stone is exposed and any mortar that was laid between the stones has been eroded. See Figure 2-27 and Figure 2-28 below.



Figure 2-27: Location 5



Figure 2-28: Location 5

The weir crest at this location is slightly higher than the central part of the weir and therefore, at low flows this part of the weir remains dry.

A photo from 1996, see Figure 2-29 below, shows this defect which doesn't appear to have significantly changed in the last 17 years.



Figure 2-29: Location 5 in 1996

Further loss of concrete capping and stone loss will lead to continuing growth of this area of damage. This defect is not considered to be a threat to the integrity of the weir in the short or medium term.

2.4.6 Other parts of the weir

Weir crest

The weir crest has a concrete capping which is generally in good condition although some sections of concrete on the upstream face below the crest are missing.

Weir glacis

Aside from the defects identified above, the weir glacis is generally in fair condition. Most of the glacis has been covered with a layer of concrete and in other areas and at the sides of the weir the original stone weir facing is visible. There are a number of transverse cracks in the glacis surface but these are not considered to be structurally significant.

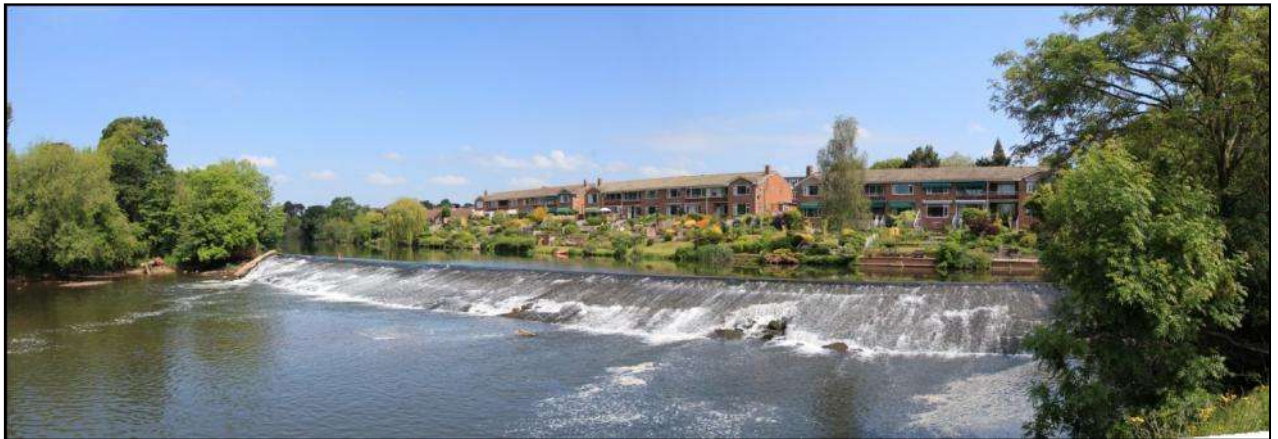


Figure 2-30: St James' Weir glacis

Weir toe

The weir toe is undercut at multiple locations due to turbulence in the weir pool. Undercutting has led to the failure of sections of the weir as described above, particularly at location 4. Undercutting was measured to be 0.3-0.5m along most of the weir toe.

Mortar loss

Where the original stonework is visible, loss of mortar from the joints is apparent allowing flow paths to form through the weir structure. This has led to the dislodging and loss of stones in some areas. As most of the weir crest and glacis has been covered by a layer of concrete, this problem isn't doesn't pose a significant threat to the weir's integrity at present. However, once several blocks have been dislodged in one area, the loss of fill material in the heart of the weir could lead to major structural damage over the long term.

2.4.7 Weir Condition Assessment Summary

A summary of the weir condition assessment is provided in Table 2-1 below

Defect Location	Time until defect results in weir breach/failure
1	> 20 years
2	> 20 years
3	10 – 20 years
4	< 10 years
5	> 20 years

Table 2-1: Weir Condition Assessment Summary

3 Options

The section describes the options that could be considered for repairing the weir and their associated costs. If the Environment Agency chooses to invest in a fish pass for the weir, it would be prudent to combine weir repairs with the construction of the fish pass.

3.1 Option 1: Do Nothing

Assuming no intervention is made, a breach in the weir is expected within 10 years (refer to Section 2.4.4 above). Assuming a breach in the weir would quickly increase in size and result in a move towards equalisation of water levels upstream and downstream, the following resulting consequences should be expected:

- Mobilisation of large amounts of fine sediment and silt that have built up behind the weir;
- Change in the tidal limit of the River Exe;
- Scour and collapse of river banks upstream of the weir due to lowered water levels;
- Drying out of St James' Mill Leat leading to significant loss of habitat and negative effect on the setting of the remains of the Mill building complex at Countess Weir;
- Negative effect on the archaeological, cultural heritage, amenity and recreational value of the area;
- Increased velocity in the river upstream of the weir would result in erosion of the left bank at the weir, eventually leading to loss of Duck's Marsh playing field land and the bridge over the Mill Leat; and
- Complaints from the public.

A positive effect of the weir breaching would be that migratory fish would be able to pass the weir much more easily than at present. This would help to improve the ecological status of the river with respect to the Water Framework Directive, at no cost to the Environment Agency.

These consequences are described in more detail in 'St James's Weir – Appraisal of Damage' report (Halcrow Group Ltd, 2012).

3.2 Option 2: Repair Weir – Do minimum

The Do Minimum option is to repair the weir to ensure its stability and integrity for the short term (i.e. at least 10 years). This requires the repair of the large hole at Location 4 described in Section 2.4.4 above.

The presence of bedrock at shallow depth at the site would make the use of sheet piling problematic as pile driving conditions would be difficult.

Therefore, it is recommended that the hole at location 4 be repaired by carrying out the following (see Figure 3-1 below):

- De-water the area using a combination of sand filled dumpy bags and a proprietary portable dam system such as 'Portadam';
- Break back concrete facing to sound material;
- Remove all loose stone, concrete and detritus from the hole;
- Excavate to bedrock at the weir toe;

- Pour a suitable fibre reinforced concrete to fill the hole and reform the weir glacis. The concrete should extend to bed rock at the weir toe to prevent scour; and
- Place graded rip rap at the weir toe.

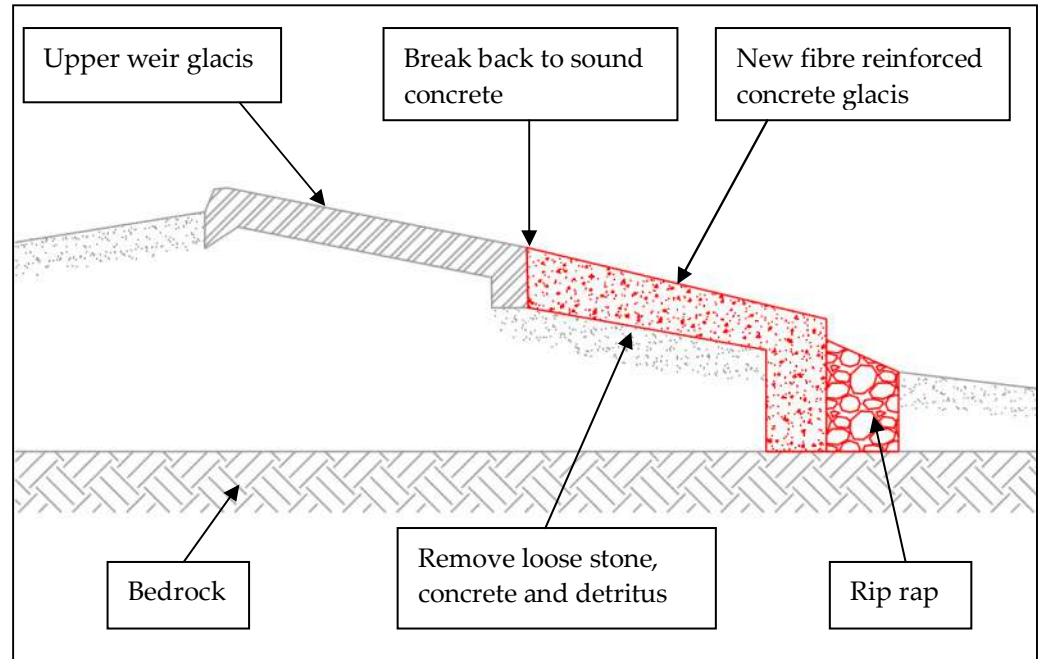


Figure 3-1: Location 4 weir repairs

The most likely access will be from the left bank. Construction traffic will reach the site across the Mill Leat Bridge. This bridge has a weight limit of 7.5 tonnes allowing access for small plant only unless temporary support/strengthening could be provided. It is envisaged that an access down to the river would be cut into the bank and a stone causeway constructed along the left bank to the weir and target repair location. Sand filled dumpy bags could be lifted onto the weir crest from the causeway. These would exclude water from the target section of the weir crest. Installation of a temporary dam system in the river bed downstream would create a dry working area allowing the repairs to be implemented. The downstream side of the weir is tidal and the works would need to be planned accordingly.

As with any in river channel construction, the work would have an associated high risk of delay as medium to high flows and high tide levels would most likely stop work.

The estimated construction cost of this option is £135k. A breakdown of the cost is provided in Appendix A. If this scheme is taken forward, it is recommended that an experienced civil engineering contractor is approached to obtain a more detailed cost estimate.

3.3 Option 3: Repair Weir – Comprehensive scheme

To repair the weir to ensure its integrity for at least the next 50 years, the following repair works are recommended:

- Repair three large holes at location 2, 3 and 4 as described in Section 3.2 above;

- Fill scour voids under the weir toe throughout the length of the toe by excavating to bedrock, filling with fibre reinforced concrete and installing rip rap erosion protection;
- Repointing of exposed stone blocks with a traditional coarse lime based mortar mix; and
- Break back areas of damaged concrete facing to sound material and patch with fibre reinforced concrete or proprietary repair mortar for smaller areas.

The estimated construction cost of this option is £430k. A breakdown of the cost is provided in Appendix A. If this scheme is taken forward, it is recommended that an experienced civil engineering contractor is approached to obtain a more detailed cost estimate

3.4 Option 4: Remove Weir

There is an option to remove the weir to improve fish passage at the site. However, the cost of this is expected to far outweigh the cost of constructing a fish pass due to the amount environmental mitigation and regrading and stabilisation of the bank and river channel required. Also, the negative effects of removing the weir described in Section 3.1 would be realised and are likely to be unacceptable to members of the public, especially locally residents.

4 Conclusion and Recommendations

St James' Weir is in a poor condition and a large breach is expected within 10 years.

A breach would ease fish passage at the weir but would however, have some significant negative impacts for the amenity, heritage, recreational and environmental value of the area which are likely to cause public concern.

Should the Environment Agency choose to invest in a fish pass for the weir, it is essential that repairs to the weir are carried out to ensure the value of the fish pass is not compromised by a failure of the weir in the short term.

As a minimum, the large hole at location 4 on the left of the weir should be repaired. This will ensure the weir's integrity for at least another 10 years. The weir could then be monitored and repairs carried out on a reactive basis. Alternatively the Environmental Agency may choose to carry out more comprehensive repairs to the weir at the same time that the fish pass is constructed.



Appendix A

Appendix A Cost Estimates

Appendix A Cost Estimates

A.1 Construction Cost Estimate for Do Minimum Option

.Item		Cost Estimate (£k)
1	Establish access route and compound, clearance of vegetation	5
2	Construct access ramp to river and causeway to Location 4	10
3	Set up and maintain temporary cofferdam/de-watering measures	25
4	Repair hole at Location 4 and install rip-rap	15
5	Reinstate and demobilise	10
6	Overheads on site works (e.g. Accommodation, fencing, track way, contractor's management and supervision costs)	5
7	Risk contingency (20% of construction cost)	14
8	Optimism Bias (60%)	50
	Total	£134k

A.2 Construction Cost Estimate for Comprehensive Repair Option

.Item		Cost Estimate (£k)
1	Establish access route and compound, clearance of vegetation	5
2	Construct access ramp to river and causeway	40
3	Set up and maintain temporary cofferdam/de-watering measures	60
4	Repair holes, fill scour holes, install rip rap	80
5	Repointing and patch repairs	10
6	Reinstate and demobilise	10
7	Overheads on site works (e.g. Accommodation, fencing, track way, contractor's management and supervision costs)	20
8	Risk contingency (20% of construction cost)	45
9	Optimism Bias (60%)	162
	Total	£432

